Independent Numerical Validation of ADR1D-ML and ADR1D-NN

Reproducible Technical Report

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Abstract

This report examines two one-dimensional reactive-transport tasks. In the first, ADR1D-ML estimates effective properties from six noisy sensor histories. In the second, ADR1D-NN computes concentration divided by inlet concentration when those properties are already known. The models and their thresholds were frozen before a challenge set of 120 new scenarios was constructed within the ADR1D domain. Each inverse scenario had five noise realizations; the neural field test comprised 299,880 position–time pairs. Samples, metrics, tolerances, and seeds were specified in advance. As an additional check, 16 scenarios were solved with finite volumes on three meshes and compared with the analytical solution. Computational cost was measured by component on one CPU thread.

ADR1D-ML met the prespecified criteria. Median absolute percentage error was 3.95 % for effective velocity and 24.94 % for effective dispersion. The classification of whether decay could be distinguished from noise achieved a balanced accuracy of 0.8357 and an area under the receiver operating characteristic (ROC) curve of 0.9113; however, 15 of 34 positive declarations were false positives. ADR1D-NN achieved a global root mean squared error (RMSE) of 0.02258 on the normalized scale and $R^2 = 0.98937$. It produced no values outside $[0, 1]$ and exactly satisfied the conditions imposed by its interface by construction. Favorable average performance coexisted with localized errors: at the worst point, the reference was zero when the source switched on at the first interior position, whereas the network predicted 0.74945.

The finite-volume solution reduced its error under refinement in all 16 cases, and every fine mesh remained within the analytical tolerance. Two cases with Pe = 350 exceeded the medium-to-fine tolerance by 6.9 % and 7.4 %; the numerical evidence is therefore retained as mixed. The analytical solution was the fastest field workload, and the timed tasks are not equivalent, so no universal speed-up is claimed. The models can be integrated into controlled experiments within ADR1D, but they were validated separately: the neural field received true parameters rather than those estimated by ADR1D-ML. This study does not demonstrate chained-workflow performance, field validation, extrapolation, or fitness for regulatory use.

Keywords: contaminant transport; machine learning; neural network; numerical validation; analytical solution; finite volumes; reproducibility.

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1 Purpose and Scope

The transport of a dissolved substance in the subsurface can be studied from two complementary directions. In the *forward problem*, transport properties are known and the change in concentration over space and time is computed. In the *inverse problem*, concentration curves are observed and the properties that produced them are recovered. In either case, favorable aggregate performance can conceal local errors: a front may, for example, arrive too early even when the average field remains close to the reference.

The purpose of this study is to determine, from verifiable evidence, which claims can be sustained about two models that represent these directions. ADR1D-ML solves the inverse problem, whereas ADR1D-NN approximates the forward problem. Both were developed from ADR1D version 1.0.0, a synthetic benchmark for one-dimensional reactive transport with parameters that remain constant within each scenario [1]. Their functions are distinct:

- **ADR1D-ML** receives six noisy sensor time histories. From these data, it estimates effective velocity, effective dispersion, whether decay can be distinguished from the noise level, and, only when that distinction is physically resolvable, a decay rate [2].
- **ADR1D-NN** receives a position, a time, the effective parameters, and the temporal definition of the source. It returns concentration divided by inlet concentration, that is, a dimensionless field value at that point [3].

This difference precludes interpreting their metrics or execution times as a competition between models: one produces scenario-level parameters and the other produces pointwise concentration values. The aim is to verify accuracy, physical consistency, regime sensitivity, reproducibility, and computational cost separately under defined input contracts. Here, *independent* means that the protocol was specified before the challenge was examined, the models were frozen, the new samples were not used for training, and the audit was performed from persisted files. It does not mean that a different physical theory was used: the challenge retains the ADR1D equation and parameter ranges.

The study is not intended to replace a site-specific hydrogeological assessment, demonstrate behavior outside the training ranges, or attribute to Water Quality Portal observations hydraulic variables that those records do not contain. The conclusions are restricted to the synthetic problem defined in this document.

The validation questions were as follows:

1. Can the published canonical results be reproduced without modifying the models or their protocols?
2. Do both models maintain acceptable performance on new scenarios that were generated before inference and do not collide with the canonical set?
3. Where are errors concentrated, and how do they differ across physical regimes, interior samples, and boundary profiles?
4. Does the analytical reference agree with an independent traditional discretization as space and time are refined?
5. What is the observed cost of each component under an explicit and reproducible measurement policy?

1.1 Reading Guide and Terminology

The report deliberately separates the physical reference, input data, model queries, and subsequent diagnostics. Table 1 summarizes terms used throughout the text.

Table 1: Operational terminology used in the validation.

Term	Meaning in this report
Base scenario	A complete combination of physical and source parameters. It is the independent unit of analysis.
Noise realization	A distinct synthetic observation of the same scenario; only the sensor noise changes.
Space–time point	An (x, t) pair within a scenario. Points on the same mesh are correlated and are not counted as independent experiments.
Analytical reference	Value computed from the closed-form ADR1D solution using the true scenario parameters.
Traditional reference	Solution obtained with the independent finite-volume method described in Section 3.4.
Frozen model	Model file identified and locked before the challenge, with no subsequent retraining or adjustment.
Active region	Points for which the normalized reference concentration is greater than 10^{-4}.
Prespecified criterion	Tolerance recorded before inference. A failure is retained and does not trigger a change in the threshold or model.

2 Physical Problem and Evaluated Artifacts

2.1 Transport Equation

ADR1D represents transport in a homogeneous medium through

$$R\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} - v\frac{\partial C}{\partial x} - \lambda RC, \tag{1}$$

where C is dissolved concentration, D is longitudinal dispersion, v is advective velocity, R is the retardation factor, and λ is the first-order decay rate. Coordinate x increases along the mean flow direction, and t is the time elapsed since the beginning of the experiment. Table 2 lists the units and interpretation.

The three terms on the right-hand side of Equation (1) have distinct roles. The second-derivative term smooths gradients through dispersion, the first-derivative term transports the profile, and $-\lambda RC$ reduces the concentration. Dividing by R defines $D_{\mathrm{eff}} = D/R$ and $v_{\mathrm{eff}} = v/R$:

$$\frac{\partial C}{\partial t} = D_{\mathrm{eff}}\frac{\partial^2 C}{\partial x^2} - v_{\mathrm{eff}}\frac{\partial C}{\partial x} - \lambda C. \tag{2}$$

This form shows which parameters the models receive: ADR1D-ML attempts to recover v_{eff} and D_{eff}, whereas ADR1D-NN receives them as known inputs. Decay retains the rate λ because R appears in both the temporal and reaction terms of Equation (1).

Scenarios with different source concentrations are compared using

$$c(x,t) = \frac{C(x,t)}{C_0}. \tag{3}$$

Table 2: Variables and parameters of the transport equation.

Symbol	Unit	Interpretation
$C(x,t)$	mg L^{-1}	Dissolved concentration at position x and time t.
v	m s^{-1}	Advective velocity before retardation is applied. It translates the center of the pulse.
D	m^2 s^{-1}	Longitudinal dispersion coefficient. It controls broadening of the front and tail.
R	Dimensionless	Retardation factor. A value greater than one reduces the effective solute velocity and dispersion relative to the reference flow.
λ	s^{-1}	First-order loss rate. In the absence of transport, concentration decreases as $\exp(-\lambda t)$.
C_0	mg L^{-1}	Concentration prescribed at the inlet while the source remains active.
t_0, τ	s	Source activation time and inlet-pulse duration.

Variable c is dimensionless. In the benchmark, its reference value is zero before arrival and one at the inlet during the pulse. An RMSE of 0.02 in this field therefore corresponds to a root mean squared error of two percentage points relative to C_0; it is not 0.02 mg L^{-1} unless $C_0 = 1$ mg L^{-1}.

The initial condition is $C(x,0) = 0$. At $x = 0$, a rectangular pulse is prescribed,

$$C(0,t) = \begin{cases} C_0, & t_0 \le t < t_0 + \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

and the analytical solution assumes decay in the far field. ADR1D evaluates a reactive extension of the Ogata–Banks-type response [4, 5]. If $u = (v^2 + 4\lambda R D)^{1/2}$, the unit-step response is

$$\begin{aligned} S(x,t) = {} & \frac{1}{2}\exp\left(\frac{(v-u)x}{2D}\right)\operatorname{erfc}\left(\frac{Rx - ut}{2\sqrt{DRt}}\right) \\ & + \frac{1}{2}\exp\left(\frac{(v+u)x}{2D}\right)\operatorname{erfc}\left(\frac{Rx + ut}{2\sqrt{DRt}}\right), \end{aligned} \tag{5}$$

for $t > 0$, and the pulse is obtained as $C(x,t) = C_0[S(x,t-t_0) - S(x,t-t_0-\tau)]$. This expression avoids associating reference labels with the error of a particular numerical mesh. Function erfc is the complementary error function. When the elapsed time supplied to S is less than or equal to zero, the response is defined as zero. Subtracting two step responses has a direct interpretation: the first switches the source on at t_0, and the second removes, after τ, the contribution of a source that would otherwise have remained active.

The closed-form solution is defined on a semi-infinite domain. The benchmark retains the first $L = 1000$ m and the first 24 h. The finite-volume reference, which requires an outlet boundary, extends the domain to 1200 m and avoids comparisons near that outlet; this distinction is detailed in Section 3.4.

The following quantities are used to interpret the design extremes:

$$\mathrm{Pe} = \frac{v_{\mathrm{eff}} L}{D_{\mathrm{eff}}}, \qquad \mathrm{Da} = \frac{\lambda L}{v_{\mathrm{eff}}}, \tag{6}$$

with $L = 1000$ m. The four regimes combine active or inactive retardation with active or zero decay: conservative, decay only, retardation only, and retardation with decay. The Peclet number compares advection and dispersion: large values correspond to narrower fronts dominated by advective transport, whereas small values correspond to more diffuse profiles. The Damköhler number compares the decay scale with the advective travel time: $\mathrm{Da} = 0$ indicates no decay, and larger values imply more visible attenuation over the travel length L.

Table 3: Physical regimes in the design. "Active" means that the parameter is sampled over its interval; "inactive" fixes $R = 1$ or $\lambda = 0$.

Regime	Retardation	Decay	Physical interpretation
Conservative	Inactive	Inactive	Transport without reactive loss or an additional reduction due to retardation.
Decay only	Inactive	Active	First-order attenuation during transport.
Retardation only	Active	Inactive	Lower effective velocity and dispersion, without reaction-driven mass loss.
Retardation and decay	Active	Active	Simultaneous retardation and attenuation.

2.2 Frozen Models

The artifacts were treated as closed boxes with public interfaces. The serialized ADR1D-ML bundle and the ADR1D-NN weights and transformations checkpoint were identified in their manifests and lock records before the challenge. Their integrity was verified before and after every operation that loaded a model. Retraining, further hyperparameter selection, changes to the classification threshold, and post hoc correction of challenge results were not permitted.

2.2.1 ADR1D-ML Inverse Workflow

Each realization supplied to ADR1D-ML contains 294 observations: 49 times at each of six sensors. The interface does not provide these 294 concentrations directly to a single estimator. It first summarizes each sensor through eleven descriptors: detected fraction, peak-to-C_0 ratio, peak time, first and last detection, normalized area, centroid, temporal spread, and cumulative times at 10, 50, and 90 % of the area. The 66 sensor descriptors, together with C_0, t_0, and τ, form 69 features per realization.

Velocity and dispersion are estimated by two independent ensembles of 300 extremely randomized trees. Both predict the base-ten logarithm of the positive parameter and use median imputation when a temporal descriptor cannot be computed, for example, because a sensor records no signal above the detection limit. The logarithmic transformation accommodates multiplicative errors and prevents negative physical predictions after conversion back to the original units.

Decay classification adds 17 descriptors that compare sensors at different distances. These include slopes of peaks, areas, centroids, and times; the quality of those fits; and physical approximations of velocity, dispersion, and attenuation. A logistic regression with 86 inputs estimates the probability that decay is resolvable. The classification truth was defined before training as

$$a_\lambda = 1 - \exp(-\mathrm{Da}), \qquad y_{\mathrm{res}} = \mathbb{I}(a_\lambda \geq 0.03)\,, \tag{7}$$

so that the physical threshold is $\mathrm{Da} \geq -\ln(0.97) = 0.0304592$. In words, decay is labeled resolvable when its ideal attenuation over one travel time is at least as large as the 3 % relative noise. This label expresses resolvability within the benchmark; it does not signify the chemical presence or absence of a reaction. A positive rate below the threshold is considered unresolvable with these sensors and this noise level.

The probability is converted to the positive class when it is greater than or equal to 0.19, a threshold selected during development and frozen before the test. Only for scenarios whose truth satisfies Equation (7) does a fourth 300-tree model estimate $\log_{10}\lambda$ from twelve selected descriptors. This separation avoids reporting a numerical estimate of λ when the expected effect falls below the defined resolution.

The five noise realizations produce five predictions per scenario. If $\widehat{z}_r > 0$ denotes a positive prediction from replicate r, the summary is their geometric mean:

$$\widehat{z}_{\mathrm{esc}} = \exp\left(\frac{1}{5}\sum_{r=1}^{5}\ln\widehat{z}_r\right). \tag{8}$$

Probabilities are averaged arithmetically, and the 0.19 threshold is applied once to that average. Each scenario thus contributes one observation to the primary metrics, while variation among replicates is retained as a diagnostic of noise sensitivity.

2.2.2 ADR1D-NN Forward Workflow

ADR1D-NN is a coordinate- and parameter-conditioned neural surrogate. Its interface receives nine columns: x, t, L, final time, v_{eff}, D_{eff}, λ, t_0, and τ. From these columns it constructs seven network inputs: normalized position and time, logarithms of effective velocity and dispersion, a $\log(1+\lambda/10^{-6})$ transformation, normalized source start time, and normalized duration. The means and standard deviations used to standardize these inputs are stored in the checkpoint.

The network is a multilayer perceptron with three hidden layers of 128 neurons, SiLU activation, a sigmoid output, and 34,177 trainable parameters. The checkpoint corresponds to epoch 57, selected with the validation set during development. The sigmoid output keeps neural predictions between zero and one. Because Equation (2) is linear in concentration, C_0 is not a network input: dimensional units are recovered as $\widehat{C} = C_0\widehat{c}$.

The interface replaces the network output with exact values on two sets: $c(x,0)=0$ for $x>0$, and the rectangular pulse at $x=0$. The order of application also resolves the corner $(0,0)$ according to the inlet condition. At all remaining interior points, including times near source activation and deactivation, the prediction comes from the network. Consequently, satisfying these two conditions verifies the implemented contract, but does not by itself demonstrate local conservation or exact satisfaction of the differential equation.

3 Validation Design

3.1 Prior Lock and Separation of Evidence

The protocol, its inputs, and the tolerances were locked before any cases were generated or new inference was executed. The initial record verified 18 frozen inputs and left the counters for new scenarios, challenge inferences, and traditional solutions at zero. Three levels of evidence were then followed:

1. **Canonical reproduction.** The previously published tests for 45 scenarios were repeated without using the challenge set.
2. **Independent challenge.** A total of 120 scenarios were generated, verified, and locked before one persisted inference per model.
3. **Traditional numerical reference.** Sixteen scenarios were selected using only true parameters and solved at three mesh levels.

The operational sequence was to (1) register the protocol, models, code, and canonical data; (2) generate the new set without loading models; (3) reconstruct it with a separate validator; (4) lock the set and its records; (5) prepare input tables without inference; (6) execute one persisted query per model; and (7) recalculate metrics from the CSV files with an auditor that neither imports the evaluator nor deserializes the models. The traditional reference was run afterward on scenarios selected solely from true parameters.

This sequence avoids two forms of optimism. First, it prevents case selection after learning where the models perform best. Second, it prevents changing a threshold or transformation to improve the result retrospectively. New metrics were not used to modify the models. A failed criterion had to be retained and discussed in the same detail as a favorable result.

3.2 Challenge Set

The design included 30 scenarios per regime: 24 interior points obtained by Latin hypercube sampling [6] and six boundary profiles. Table 4 summarizes the ranges. Positive variables spanning several orders of magnitude were sampled on a logarithmic scale; the others were sampled linearly. In Latin hypercube sampling, each transformed interval is divided into 24 strata, and each stratum is used once per dimension. This improves interval coverage with fewer points than a Cartesian grid, but does not turn the scenarios into a sample of naturally occurring aquifer frequencies.

Table 4: Ranges of the locked challenge set.

Variable	Range	Design transformation
Velocity v	0.05–0.35 m s^{-1}	Linear
Dispersion D	1–20 m^2 s^{-1}	$\log_{10}$
Source concentration C_0	0.2–5 mg L^{-1}	$\log_{10}$
Source start t_0	0–3600 s	Linear
Duration τ	1800–10800 s	Linear
Active retardation R	1.1–3.0	Linear
Active decay λ	10^{-7}–2×10^{-5} s^{-1}	$\log_{10}$

The six boundary profiles per regime are not random observations. The first four combine the minimum and maximum velocity and dispersion values while holding all other transformed coordinates at their midpoints. The remaining two combine opposing extremes of concentration, start time, duration, retardation, and decay. Their purpose is to expose the model to documented corners of the domain that open interior sampling does not reach exactly. When a process is inactive in a regime, $R = 1$ or $\lambda = 0$ is fixed.

The neural network receives effective parameters. Because of division by R, the covered effective range is approximately 0.0167–0.35 m s^{-1} for velocity and 0.333–20 m^2 s^{-1} for dispersion. Every challenge case remains within the strict ADR1D-ML and ADR1D-NN input contracts; the boundary profiles test permitted extremes, not extrapolation.

Each scenario field contains 51 positions between 0 and 1000 m and 49 times between 0 and 86400 s, for a total of 299,880 values. ADR1D-ML used six sensors at 100, 250, 400, 600, 800, and 1000 m. Each scenario received five independent realizations of 3 % relative Gaussian noise, an absolute floor of 0.002 mg L^{-1}, a detection limit of 0.01 mg L^{-1}, and substitution of 0.005 mg L^{-1} below that limit. This produced 600 realizations and 176,400 observations.

For each observation with truth C_{ref}, the following model was used:

$$\sigma(C_{\mathrm{ref}}) = \max\Big(0.002 \text{ mg L}^{-1}, 0.03C_{\mathrm{ref}}\Big)\,, \tag{9}$$

$$C_{\mathrm{uncensored}} = \max(0, C_{\mathrm{ref}} + \sigma Z)\,, \qquad Z \sim \mathcal{N}(0,1). \tag{10}$$

If $C_{\mathrm{uncensored}} < 0.01$ mg L^{-1}, the value supplied to the feature workflow was 0.005 mg L^{-1} and was marked as not detected. During extraction, that mark preserves the frequency of nondetections but assigns zero to the integrated signal. This convention separates the value used to represent a report below the detection limit from the value used to compute area, centroid, and cumulative times.

The parameter seed was 2026072105, the noise seed was 2026072106, and the bootstrap seed was 2026072107. The validator reconstructed the scenarios, fields, true concentrations, and every noise sequence without loading the models. The maximum absolute difference from the reconstruction was approximately 5.0×10^{-12} mg L^{-1}. There were no duplicate identifiers or parameters relative to the 300 canonical scenarios; the minimum normalized distance was 0.1170. A second temporary generation produced files that were identical byte for byte. The distance was computed after mapping each parameter to its normalized linear or logarithmic coordinate. Thus, 0.1170 indicates separation in design space; it is neither a physical distance in meters nor a measure of similarity between complete fields.

3.3 Metrics and Statistical Analysis

3.3.1 Regression and Field Errors

Let y_i be a reference, $\widehat{y}_i$ its prediction, and $e_i = \widehat{y}_i - y_i$. The basic metrics were computed as

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |e_i|, \tag{11}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2}, \tag{12}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} e_i^2}{\sum_{i=1}^{N}(y_i - \overline{y})^2}. \tag{13}$$

Mean absolute error (MAE) expresses the average magnitude of error without regard to sign. Root mean squared error (RMSE) gives greater weight to large errors because it squares the residuals. Coefficient R^2 compares model error with the error obtained by always predicting the mean reference value: one indicates perfect agreement; zero indicates no improvement over that mean; and a negative value indicates worse performance. A high R^2 does not guarantee a small maximum error or correct front arrival.

For positive parameters spanning several orders of magnitude, the same equations were applied to $\log_{10} y_i$ and $\log_{10} \widehat{y}_i$. Absolute percentage error (APE) was also computed as

$$\mathrm{APE}_i = 100\frac{|\widehat{y}_i - y_i|}{y_i}. \tag{14}$$

Median APE describes a typical scenario, and the 90th percentile indicates the value below which 90 % of scenarios fall. A 90th percentile of 65 %, for example, does not mean that every error is 65 %; it means that approximately one tenth are larger. ADR1D-ML uses one aggregate prediction for each of the 120 scenarios. The conditional evaluation of λ uses only the 23 scenarios whose truth is resolvable.

ADR1D-NN was summarized in two ways. *Pooled* metrics use all 299,880 points to describe the complete field. *Per-scenario* metrics are first computed on each 2499-point mesh and then summarize the distribution of 120 RMSE values. This second view prevents a scenario with a simple or nearly zero field from determining the conclusion by itself.

3.3.2 Resolvability Classification

The confusion matrix distinguishes true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The following metrics were used:

$$\mathrm{Sens} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad \mathrm{Spec} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}, \tag{15}$$

$$\mathrm{BA} = \frac{1}{2}(\mathrm{Sens}+\mathrm{Spec}), \qquad \mathrm{Prec} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \tag{16}$$

$$F_1 = 2\frac{\mathrm{Prec}\ \mathrm{Sens}}{\mathrm{Prec}+\mathrm{Sens}}. \tag{17}$$

Sensitivity answers how many resolvable cases were recovered; precision answers how many positive declarations were correct. Balanced accuracy averages sensitivity and specificity so that the minority class is not hidden. The area under the ROC curve evaluates probability ranking across all thresholds, whereas the confusion matrix and F_1 depend on the frozen 0.19 threshold. Log loss penalizes confident probabilities assigned to the wrong class. These metrics are not interchangeable: a classifier may rank cases well while producing too many false positives at a particular threshold.

3.3.3 Physical Diagnostics of the Neural Field

The active region was defined by $c_{\mathrm{ref}} > 10^{-4}$. MAE and RMSE were reported for both the entire field and this region because broad areas with a zero or nearly zero reference can artificially lower the global average. The initial condition, inlet boundary, and fractions of predictions below zero or above one were also verified.

For each scenario, the spatial integral was first computed as

$$Q(t) = \int_0^L c(x,t)\,dx, \tag{18}$$

and then $\mathcal{M} = \int_0^T Q(t)\,dt$. The diagnostic termed *relative integrated-mass error* is $|\widehat{\mathcal{M}} - \mathcal{M}|/\mathcal{M}$. Because c is normalized and the model does not include cross-sectional area, bulk density, or porosity, this quantity is a comparative field integral rather than the total dimensional mass of contaminant in an aquifer.

At each interior position where the reference reaches 10^{-4}, arrival time was defined as the first tabulated time above that threshold. The absolute error between the first reference and predicted arrivals was computed and summarized per scenario by the spatial median. If the prediction never crossed the threshold, the final time plus one time step was assigned as a penalty. Because the field is stored every 1800 s, this diagnostic is quantized in 1800 s intervals.

Finally, the following residual was approximated at interior nodes:

$$r = \frac{\partial \widehat{c}}{\partial t} - D_{\mathrm{eff}}\frac{\partial^2 \widehat{c}}{\partial x^2} + v_{\mathrm{eff}}\frac{\partial \widehat{c}}{\partial x} + \lambda \widehat{c} \tag{19}$$

using centered differences, and its RMSE was reported. This residual depends on the diagnostic mesh, amplifies abrupt changes, and is not written in conservative form; it was therefore recorded without an acceptance threshold.

3.3.4 Independent Unit and Intervals

The resampling unit was always the base scenario. The 95 % intervals were computed from 5000 replicates of grouped percentile bootstrap resampling [8]. In each replicate, scenario identifiers were sampled with replacement, and their five noise realizations or 2499 points were kept together. Treating every point as an independent observation would have produced artificially narrow intervals because neighboring values belong to the same solution and are correlated.

The minimum criteria were specified before the challenge was examined. They were chosen as sufficiency tolerances for deciding whether the models could continue to be used in ADR1D experiments, not as regulatory limits or guarantees for a real case. The intervals describe variation among scenarios in the design; they are not predictive intervals for an individual scenario.

3.4 Traditional Numerical Reference

The traditional reference was implemented separately from the analytical generation and ADR1D-NN. Its purpose was not to compete with the network in execution time, but to answer two questions: whether the closed-form solution is consistent with a conservative discretization and whether error decreases as space and time are refined.

Equation (2) can be written as

$$\frac{\partial c}{\partial t} + \frac{\partial J}{\partial x} = -\lambda c, \qquad J = v_{\text{eff}} c - D_{\text{eff}} \frac{\partial c}{\partial x}. \tag{20}$$

The domain was divided into volumes of uniform width Δx, and an average concentration was stored at the center of each cell. At a face between cells i and $i+1$, the Scharfetter–Gummel flux was [7]

$$J_{i+1/2} = \frac{D_{\text{eff}}}{\Delta x} \left[B(-P_h) c_i - B(P_h) c_{i+1} \right], \qquad P_h = \frac{v_{\text{eff}} \Delta x}{D_{\text{eff}}}, \tag{21}$$

where $B(z) = z/(\exp z - 1)$ and $B(0) = 1$. This fitting reproduces centered diffusion when P_h is small and avoids the negative oscillations that a conventional centered approximation can introduce when advection dominates. The identity $B(-P_h) - B(P_h) = P_h$ ensures that a constant state carries the correct advective flux.

After the fluxes were assembled, the semidiscrete system took the form $d\mathbf{c}/dt = A\mathbf{c} + \mathbf{b}c_{\text{in}}(t)$. Each step was solved with backward Euler:

$$(I - \Delta t A)\mathbf{c}^{n+1} = \mathbf{c}^n + \Delta t\, \mathbf{b}\, c_{\text{in}}(t_{n+1/2}). \tag{22}$$

The method is implicit and does not require the stability restriction of an explicit Euler scheme; its accuracy nevertheless depends on Δt. Factorizations were reused whenever the same time step recurred. Times t_0 and $t_0 + \tau$ were inserted exactly into the time grid so that no step crossed a source change without representing it.

The computational domain was extended to 1200 m. At the inlet, the Dirichlet pulse was imposed over a half-cell distance. At the outlet, the diffusive flux was set to zero and the advective flux $v_{\text{eff}} c$ was allowed. The comparison was restricted to $0 \le x \le 900$ m to preserve at least 300 m between reported points and this artificial boundary.

Four scenarios were selected per regime: minimum Peclet, maximum Peclet, maximum Damköhler or greatest advective travel time when $\lambda = 0$, and the medoid of normalized parameter space. The medoid is the existing scenario with the smallest total distance to the other points in its regime; unlike a centroid, it always corresponds to an actual case in the set. Selection did not consult model or solver errors. The three discretizations were

Level	Δx	Maximum Δt
Coarse	10 m	225 s
Medium	5 m	112.5 s
Fine	2.5 m	56.25 s

The three meshes are nested. To compare levels, the averages of two adjacent fine cells were restricted to one cell at the next level; displaced centers were not compared through arbitrary interpolation. The analytical solution was evaluated at cell centers and at the 49 reporting times. The empirical order between successive errors was computed as

$$p = \frac{\log(E_h/E_{h/2})}{\log 2}. \tag{23}$$

Because Δx and the maximum time step were reduced simultaneously, p describes the joint space–time campaign rather than the isolated asymptotic order of one discretization.

Balance was checked at every step using the change in the discrete integral, the outlet flux, inlet flux, and reactive loss:

$$r_{\mathrm{bal}} = \frac{M^{n+1} - M^n}{\Delta t} + J_{\mathrm{out}} - J_{\mathrm{in}} + \lambda M^{n+1}, \qquad M^n = \Delta x \sum_i c_i^n. \tag{24}$$

Acceptance required, in each of the 16 cases, a fine-mesh RMSE relative to the analytical solution no greater than 0.03, an RMSE between the restricted medium and fine meshes no greater than 0.01, and a minimum concentration no lower than -10^{-10}. Discrete balance, maximum error, arrival time, and observed order were retained as diagnostics. The first criterion measures agreement with a reference external to the mesh; the second asks whether two successive resolutions are already sufficiently close under the adopted tolerance.

3.5 Computational Cost

The benchmark was run on a CPU with one computational thread. Each workload had two warm-ups and seven repetitions measured with `time.perf_counter_ns`; median, interquartile range, minimum, and maximum were reported. Data were loaded into memory before the computational kernels were timed, so CSV reading and output serialization were excluded.

Separation by component prevents assigning the entire cost to the model. Loading includes verifying and deserializing the artifact; feature construction transforms data already resident in memory; inference calls the frozen estimator; and the end-to-end path combines these three steps while continuing to exclude file input and output. For the traditional reference, analytical evaluation, finite-volume integration, and the workflow that prepares the 16 cases and executes both calculations were timed separately.

The two warm-ups initialize deferred imports, caches, and internal paths before times are summarized. All seven measurements are retained in the JSON so that the median does not conceal their dispersion. The interquartile range is the difference between the 75th and 25th percentiles and describes the central half of the repetitions.

To avoid turning timing into a second query of the challenge set, model times were measured only on the published canonical partition: 45 scenarios for ADR1D-ML and 112,455 points for ADR1D-NN. The analytical evaluation and traditional solver used the 16 selected cases on the fine mesh, with 376,320 retained values. Every repetition produced the same output. A separate probe with `tracemalloc` described allocations visible from Python; it is not interpreted as total resident memory because it may omit allocations made by native libraries.

4 Results

4.1 Reproduction and Integrity

ADR1D-ML reproduced five prediction arrays and four metric checks over its 45 canonical scenarios. ADR1D-NN reproduced 112,455 predictions, 60 scalar metrics, 45 per-scenario rows, four diagnostic selections, and 4,455 exact physical constraints. In both cases, the reconstructed report was identical to the published report byte for byte.

Binary identity is a more demanding test than observing similar figures: it confirms that, with the same inputs and environment, record order, serialized precision, and the full report content were preserved. This test establishes reproducibility of the canonical evaluation; it does not replace the new test because those 45 scenarios had already been part of the development history.

Challenge inference was executed once per model after locking. No subsequent adjustment was made. An audit that neither imported the evaluation modules nor loaded models recalculated the evidence from the CSV files: it completed 2664 checks, compared 2423 values, and found a maximum absolute difference of 8.92×10^{-13}, attributable to serialization precision. The protected models, protocols, inputs, and results remained unchanged.

The 2664 checks included counts, identifiers, global metrics, per-scenario results, bootstrap intervals, physical constraints, and integrity checks. The maximum difference is approximately eleven orders of magnitude smaller than the model RMSE and therefore reflects file rounding rather than a material discrepancy between evaluator and auditor.

4.2 Parameter Estimation with ADR1D-ML

Figure 1 shows scenario-level aggregation. Velocity was recovered with less relative dispersion than D_{eff} and λ. Across the 120 scenarios, median absolute percentage error for velocity was 3.95 %, its 90th percentile was 10.87 %, and logarithmic R^2 was 0.9892. The corresponding values for dispersion were 24.94 %, 65.81 %, and 0.8494.

In distributional terms, half of the velocity estimates deviated by less than 3.95 %, and nine out of ten deviated by less than 10.87 %. For dispersion, half remained below 24.94 %, but approximately one tenth exceeded 65.81 %. Meeting the dispersion tolerance should not be read as uniform accuracy: the protocol allowed a larger error for a property that is more difficult to identify from the broadening of censored curves.

The classification produced 82 true negatives, 15 false positives, four false negatives, and 19 true positives. Sensitivity was 0.8261, but precision was 0.5588. Specificity was 0.8454, and its average with sensitivity gives the balanced accuracy of 0.8357. In other words, the model recovered 19 of the 23 resolvable cases, but only 19 of its 34 positive declarations were correct. This behavior may be reasonable for a screening stage if a false negative is more costly, but that preference was not part of the protocol and should not be assumed for a specific decision. A positive class indicates that decay merits further examination; it does not confirm decay by itself.

The conditional regression was evaluated on 23 scenarios, above the minimum of ten specified for applying the criterion. The 95 % bootstrap interval for its median APE was 16.59–38.93 %.

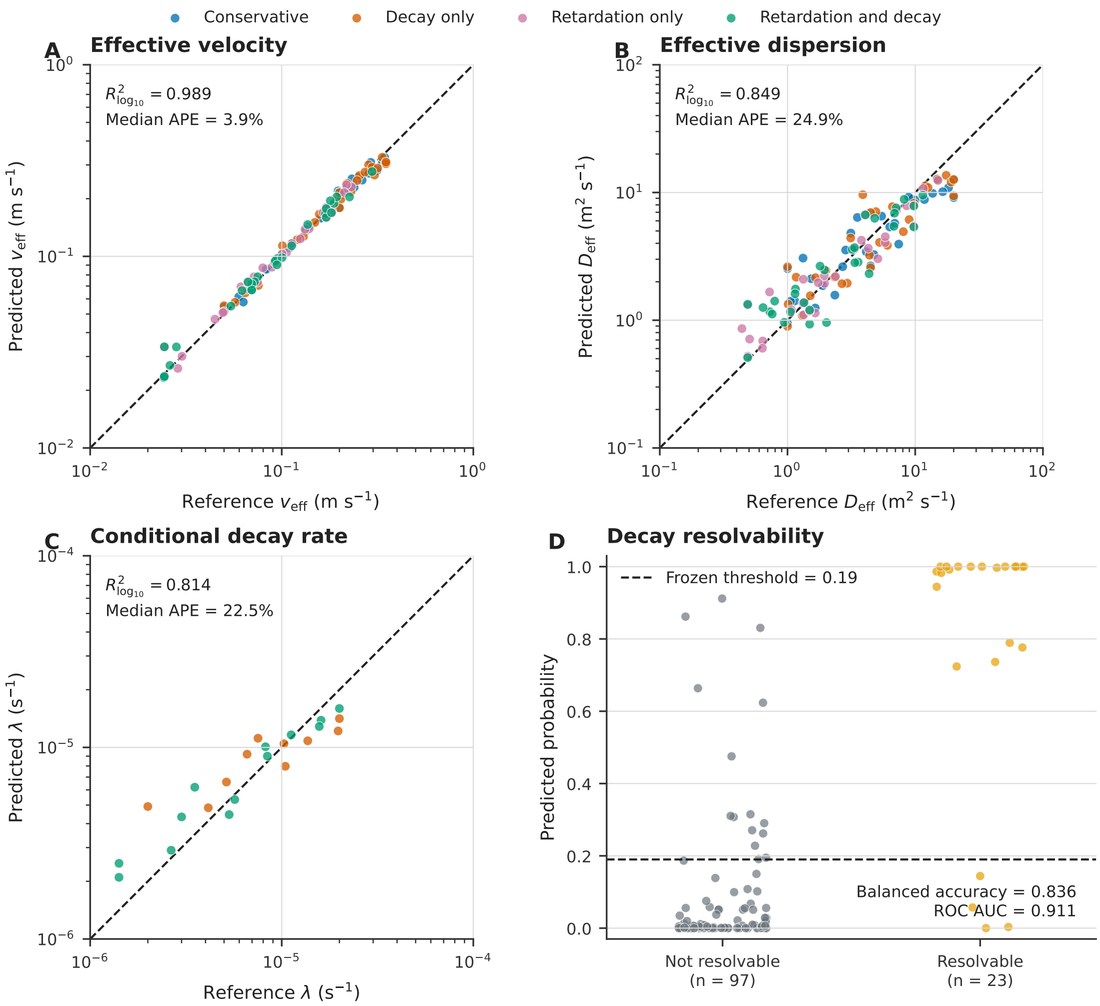

Figure 1: Validation of ADR1D-ML on 120 new scenarios. (A–C) Agreement between the reference and aggregate prediction; the dashed line indicates identity. (D) Probability of decay resolvability and the frozen threshold. Each point represents one base scenario, not an individual noise realization.

The corresponding intervals were 0.0255–0.0395 for logarithmic velocity RMSE, 0.1524–0.1962 for logarithmic dispersion RMSE, 0.7418–0.9115 for balanced accuracy, and 0.8089–0.9856 for ROC area.

Noise affected the outputs unevenly. Median coefficient of variation among replicates was 0.66 % for velocity, 2.82 % for dispersion, and 6.21 % for conditional decay. The 90th percentile of probability standard deviation was 0.274, even though its median was 0.007. Overall classification stability does not therefore imply equal probability stability near the threshold. All five individual classes agreed in 96 of the 120 scenarios; in the remaining 24, at least one replicate fell on the opposite side of 0.19. Retaining all five probabilities reveals sensitivity concealed by a single aggregate label.

Across regimes, median dispersion APE ranged from 16.22 % for retardation only to 30.73 % for decay only; logarithmic R^2 remained between 0.806 and 0.882. Velocity was more uniform, with median APE values from 3.31 % to 4.07 %. The evidence supports velocity estimation more strongly than an interchangeable reading of all outputs. In particular, velocity, dispersion, and decay should not be propagated into a later simulation as though they shared the same relative error.

Table 5: Primary ADR1D-ML criteria on the locked challenge.

Metric	Observed	Criterion	Assessment
Median APE of v_{eff}	3.95 %	≤ 10 %	**Pass**
90th-percentile APE of v_{eff}	10.87 %	≤ 25 %	**Pass**
$R^2_{\log_{10}}$ of v_{eff}	0.9892	≥ 0.90	**Pass**
Median APE of D_{eff}	24.94 %	≤ 40 %	**Pass**
90th-percentile APE of D_{eff}	65.81 %	≤ 100 %	**Pass**
$R^2_{\log_{10}}$ of D_{eff}	0.8494	≥ 0.50	**Pass**
Balanced accuracy of resolvability	0.8357	≥ 0.65	**Pass**
ROC area of resolvability	0.9113	≥ 0.70	**Pass**
Median APE of conditional λ	22.51 %	≤ 85 %	**Pass**
$R^2_{\log_{10}}$ of conditional λ	0.8139	≥ 0.10	**Pass**

4.3 Field Prediction with ADR1D-NN

Across 299,880 points, ADR1D-NN achieved an MAE of 0.00629, an RMSE of 0.02258, and $R^2 = 0.98937$. In the active region, RMSE increased to 0.03887. Median per-scenario RMSE was 0.01923, the 90th percentile was 0.03100, and the maximum was 0.05323. Table 6 confirms that all 12 specified criteria were met.

Because the output is $c = C/C_0$, the global RMSE corresponds to 2.26 percentage points of inlet concentration and the active RMSE to 3.89 points. Of the 299,880 points, 79,556 (26.5 %) belonged to the active region and 220,324 (73.5 %) to the inactive region. This asymmetry explains why an active-region metric is needed in addition to the global average.

Table 6: Primary ADR1D-NN criteria on the locked challenge.

Metric	Observed	Criterion	Assessment
Global RMSE	0.02258	≤ 0.035	**Pass**
Active-region RMSE	0.03887	≤ 0.065	**Pass**
Global R^2	0.98937	≥ 0.97	**Pass**
90th percentile of per-scenario RMSE	0.03100	≤ 0.05	**Pass**
Median relative mass error	1.74 %	≤ 10 %	**Pass**
90th percentile of relative mass error	7.82 %	≤ 25 %	**Pass**
Median arrival-time error	1800 s	≤ 1800 s	**Pass**
90th percentile of arrival-time error	1800 s	≤ 3600 s	**Pass**
Negative predictions	0	$= 0$	**Pass**
Predictions greater than one	0	$= 0$	**Pass**
Maximum initial-condition error	0	$\leq 10^{-12}$	**Pass**
Maximum inlet-boundary error	0	$\leq 10^{-12}$	**Pass**

The 95 % bootstrap intervals were 0.02081–0.02447 for global RMSE, 0.03747–0.04047 for active RMSE, and 0.02630–0.03347 for the 90th percentile of per-scenario RMSE. Mass error was favorable in most cases, whereas arrival-time error was concentrated at 1800 s, equal to the temporal interval of the evaluation table. The latter result must be read at the available resolution: it does not demonstrate subinterval accuracy. Median relative error of the space–time integral was 1.74 %; it did not exceed 7.82 % in 90 % of scenarios. This indicates good agreement of integrated content but does not guarantee that the content is located at the same position or arrives at the same time.

Case `A5-CH-0056` belongs to the decay-only regime, was a boundary profile, and had $\mathrm{Pe} = 2.5$. Its

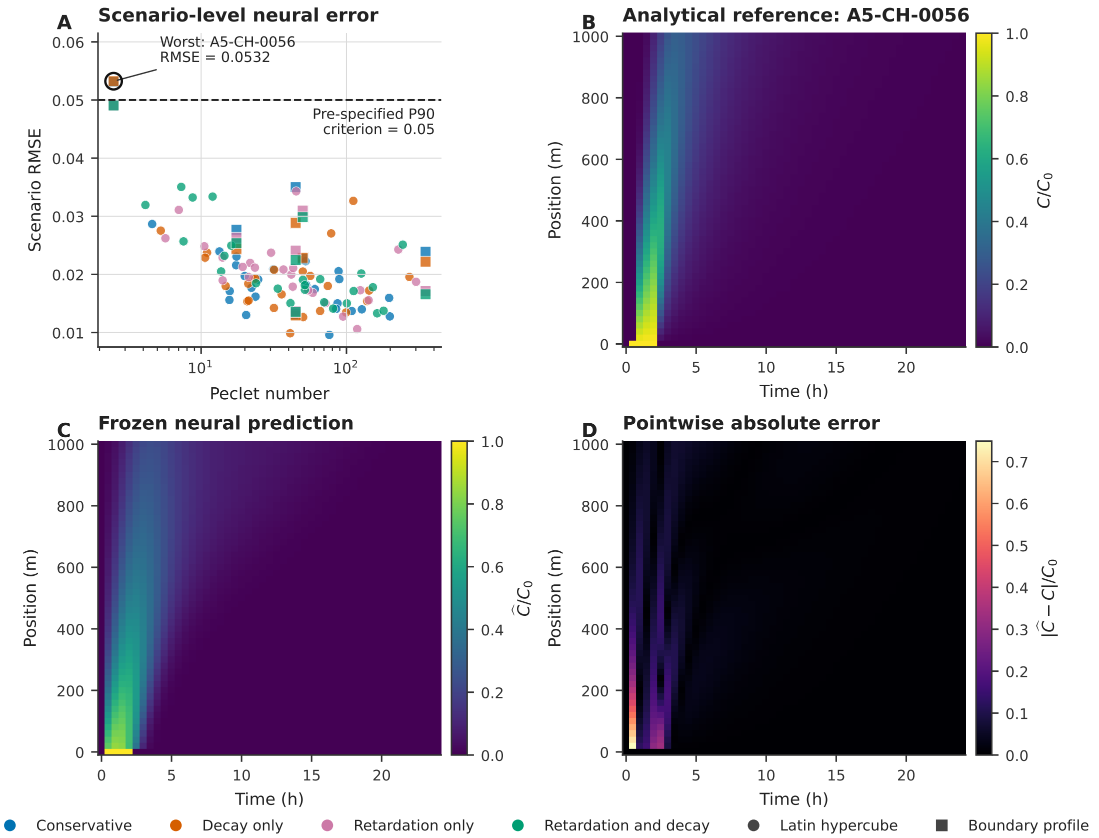

Figure 2: Heterogeneity of ADR1D-NN error. (A) Per-scenario RMSE versus Peclet number; the criterion applies to the 90th percentile, not the individual maximum. (B–D) Analytical reference, frozen prediction, and absolute error for the scenario with the largest persisted RMSE, `A5-CH-0056`. Squares identify boundary profiles and circles identify Latin hypercube points.

RMSE was 0.05323, $R^2 = 0.87276$, active RMSE was 0.03662, and maximum error was 0.74876. Figure 2 shows that the difference is concentrated around the early evolution of the pulse and its front rather than spread uniformly across the 24 h. The maximum pointwise error in the entire challenge, 0.74945, occurred in another boundary profile with Peclet 2.5, `A5-CH-0026`. The coexistence of a low global RMSE and large maximum errors is explained in part by the large proportion of inactive or near-zero cells.

The two maxima locate the mechanism. In `A5-CH-0056` and `A5-CH-0026`, the source starts at $t = 1800$ s, the first interior position is $x = 20$ m, and the reference remains zero at the exact activation time. The network predicted 0.74876 and 0.74945, respectively. The point is not covered by the hard constraints because it belongs to neither $t = 0$ nor $x = 0$. This is an early interior arrival at a temporal discontinuity, not a small deviation distributed throughout the field. Moreover, because the reference is less than 10^{-4}, these maxima belong by definition to the inactive region and do not appear in the active maximum.

Boundary profiles had a pooled RMSE of 0.02995, compared with 0.02032 at Latin hypercube points. The lowest Peclet quartile had a pooled RMSE of 0.03033, the largest among quartiles. Active RMSE, however, was largest in the highest Peclet quartile (0.05117), where the front occupies fewer points and active errors have less weight in the global average. These observations support examining both active regions and complete profiles. They also show that “higher Peclet” does not automatically imply a higher global RMSE: the upper quartile has fewer active cells, so the total average is dominated by zeros even though error within the front is larger.

No values fell outside $[0, 1]$. The discrete PDE residual was 1.961×10^{-4} s^{-1}. It is reported as a diagnostic: it depends on the discretization used to compute derivatives and does not constitute a test of local conservation or exact satisfaction of the equation. Its value is not compared directly with λ because the residual combines temporal, advective, dispersive, and reactive derivatives, and centered differences are particularly sensitive to abrupt pulse changes.

4.4 Numerical Reference and Refinement

The local analytical implementation was first checked against 39,200 values from the locked field; the maximum absolute difference was 6.50×10^{-12}. The 16 selected cases were solved at three levels, for a total of 48 runs. RMSE relative to the analytical solution decreased monotonically in every case.

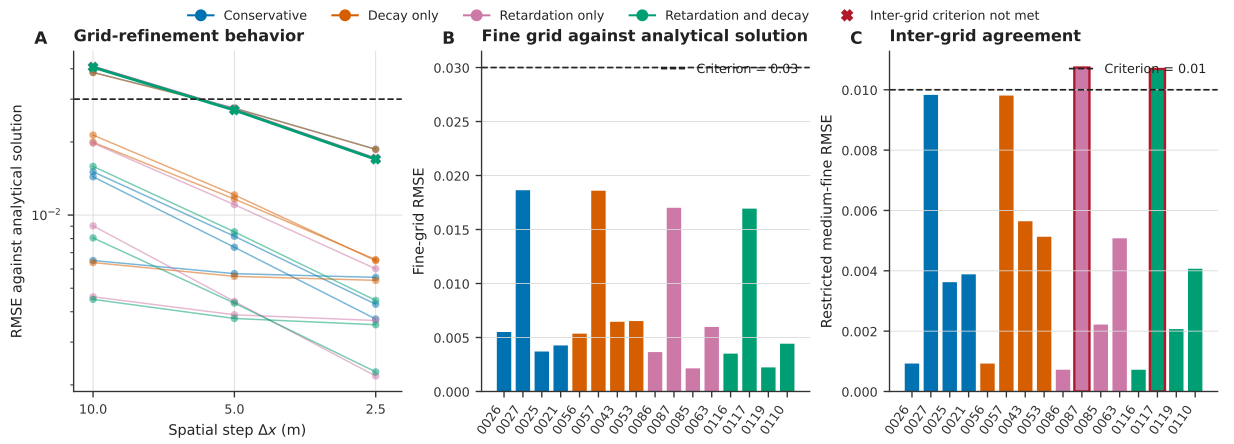

Figure 3: Traditional reference. (A) RMSE relative to the analytical solution under space and time refinement. (B) Fine-mesh RMSE in the 16 cases. (C) RMSE between medium and fine meshes restricted to common points. Red outlines mark the two cases that exceeded the intermesh tolerance.

Table 7: Summary of the traditional numerical reference.

Diagnostic	Observed	Criterion	Assessment
Median fine-mesh RMSE	0.00546	≤ 0.03 per case	**Pass**
Maximum fine-mesh RMSE	0.01867	≤ 0.03 per case	**Pass**
Median medium–fine RMSE	0.00399	≤ 0.01 per case	—
Maximum medium–fine RMSE	0.01074	≤ 0.01 per case	**Mixed evidence**
Minimum normalized concentration	0.0	$\geq -10^{-10}$	**Pass**
Maximum absolute balance residual	4.45×10^{-13}	Diagnostic	—

Table 8 expands the aggregate result and permits review of all 16 cases without relying on the figure. "Medium–fine" compares cell averages on the same restricted mesh; the two bold values are the only ones above 0.01.

Fourteen scenarios met every criterion. `A5-CH-0117` (retardation with decay) and `A5-CH-0087` (retardation only), both selected for maximum Peclet and both with $\mathrm{Pe} = 350$, produced medium–fine RMSE values of 0.01069 and 0.01074. The exceedances above 0.01 were small but were retained without relaxing the threshold. Both cases did meet the fine-mesh tolerance relative to the analytical solution, with RMSE values of 0.01697 and 0.01704.

The medium–fine exceedances were 6.9 % and 7.4 % relative to the tolerance, not orders of magnitude. Their coincidence at $\mathrm{Pe} = 350$ nevertheless has a concrete numerical explanation. Cell

Table 8: Individual results from the traditional reference. Pe and Da are computed from the true parameters.

Case	Selection criterion	Pe	Da	Fine RMSE	Medium–fine RMSE
Conservative					
`A5-CH-0026`	Minimum Pe	2.50	0	0.00554	0.00094
`A5-CH-0027`	Maximum Pe	350.00	0	0.01867	0.00984
`A5-CH-0025`	Maximum advective travel time	50.00	0	0.00373	0.00363
`A5-CH-0021`	Medoid	31.48	0	0.00430	0.00389
Decay only					
`A5-CH-0056`	Minimum Pe	2.50	0.02828	0.00539	0.00094
`A5-CH-0057`	Maximum Pe	350.00	0.00404	0.01863	0.00982
`A5-CH-0043`	Maximum Da	74.42	0.13230	0.00649	0.00565
`A5-CH-0053`	Medoid	50.33	0.00441	0.00655	0.00514
Retardation and decay					
`A5-CH-0116`	Minimum Pe	2.50	0.05798	0.00354	0.00073
`A5-CH-0117`	Maximum Pe	350.00	0.00828	0.01697	**0.01069**
`A5-CH-0119`	Maximum Da	44.72	0.30000	0.00226	0.00208
`A5-CH-0110`	Medoid	51.73	0.01751	0.00445	0.00408
Retardation only					
`A5-CH-0086`	Minimum Pe	2.50	0	0.00368	0.00073
`A5-CH-0087`	Maximum Pe	350.00	0	0.01704	**0.01074**
`A5-CH-0085`	Maximum advective travel time	50.00	0	0.00218	0.00223
`A5-CH-0063`	Medoid	58.58	0	0.00601	0.00509

Peclet is $P_h = \mathrm{Pe}\,\Delta x/L$: it is 1.75 on the medium mesh and 0.875 on the fine mesh. Both solutions remain stable and nonnegative because of exponential fitting, but they represent a narrow front at different resolutions. The intermesh criterion detects that displacement even though both solutions maintain an RMSE below 0.03 relative to the analytical solution.

The largest fine-mesh RMSE, 0.01867, corresponded to the conservative case `A5-CH-0027`, also with $\mathrm{Pe} = 350$. This pattern is consistent with the greater difficulty of resolving advection-dominated fronts at fixed resolution. Median observed order was 0.681 from coarse to medium and 0.751 from medium to fine. These values describe the joint campaign in Δx and Δt; they are not a formal asymptotic estimate of scheme order.

All concentrations were nonnegative, and the maximum absolute residual of the discrete balance was 4.45×10^{-13}; the maximum relative residual was 2.72×10^{-9}. The evidence supports the consistency of the reference in the examined cases, but the two intermesh failures justify the aggregate result of *mixed evidence* and support further refinement when Peclet approaches the upper end of the domain.

4.5 Computational Cost

Figure 4 and Table 9 present the medians of seven repetitions. Interquartile ranges were narrow relative to the medians, and each workload produced an identical output in the two warm-ups, the seven measured repetitions, and the separate memory probe.

Pure ADR1D-ML inference occupied a smaller fraction of total time; feature construction and verification/loading contributed more. The opposite occurred for ADR1D-NN: for 112,455 points, inference dominated the time and input construction was brief. In the traditional reference, the numerical solution dominated analytical evaluation.

Each row represents a complete batch, not the fixed cost of an isolated prediction. Throughput in scenarios or points per second is obtained by dividing batch cardinality by median time and may change with a different batch size. In a service that keeps the model in memory, loading cost

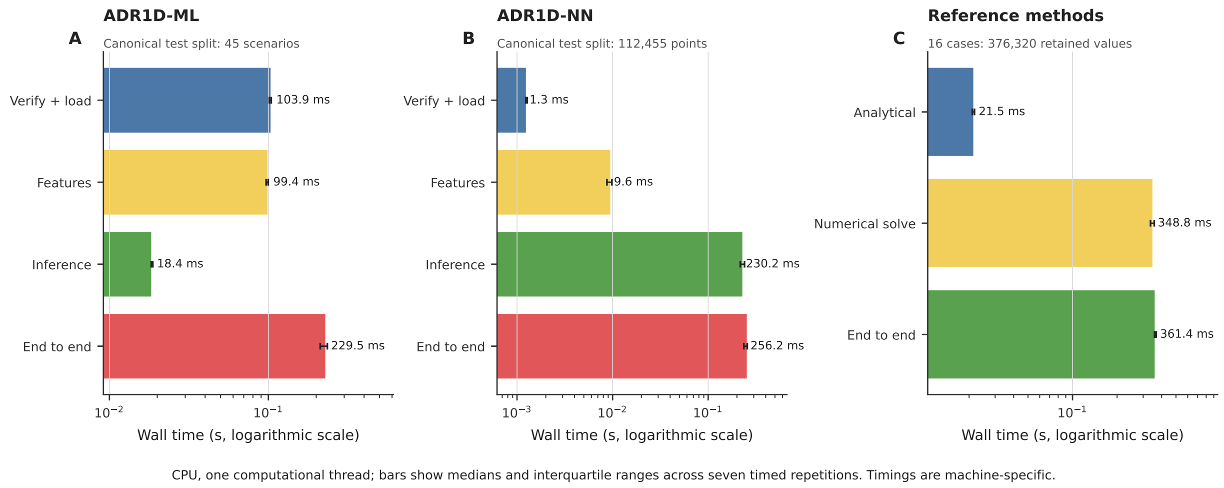

Figure 4: Computational cost by component. Bars indicate the median and interquartile range. Each panel has a different workload and should not be used to compute a direct speed-up between models or methods.

Table 9: Wall-clock times on a single CPU thread. IQR: interquartile range.

Workload	Median (ms)	IQR (ms)	Median throughput
ADR1D-ML: verify and load bundle	103.87	2.90	9.63 loads s^{-1}
ADR1D-ML: construct 45 rows	99.41	3.02	452.7 scenarios s^{-1}
ADR1D-ML: infer 45 scenarios	18.44	0.42	2441.0 scenarios s^{-1}
ADR1D-ML: end to end	229.54	23.17	196.0 scenarios s^{-1}
ADR1D-NN: verify and load network	1.25	0.04	799.5 loads s^{-1}
ADR1D-NN: construct 112,455 rows	9.55	1.11	1.177×10^7 points s^{-1}
ADR1D-NN: infer 112,455 points	230.24	23.24	4.884×10^5 points s^{-1}
ADR1D-NN: end to end	256.23	20.80	4.389×10^5 points s^{-1}
Analytical reference: 376,320 values	21.50	0.75	1.750×10^7 values s^{-1}
Traditional solution: 376,320 values	348.84	20.20	1.079×10^6 values s^{-1}
Numerical reference, end to end	361.36	9.27	1.041×10^6 pairs s^{-1}

would be amortized; in a command-line execution started for every query, it would be paid again. This study reports both components but does not simulate a persistent service.

These figures describe an Apple arm64 computer with macOS 26.5.2, 10 logical CPUs, 24 GiB of memory, and CPython 3.12.2. The environment used NumPy 2.0.2, pandas 2.2.3, SciPy 1.15.2, scikit-learn 1.7.0, joblib 1.4.2, and PyTorch 2.7.1. The workloads are not equivalent: ADR1D-ML solves one inverse problem per scenario, ADR1D-NN evaluates a pointwise surrogate, and the numerical reference retains values from 16 cases on a different mesh. A speed-up factor is therefore not computed.

The analytical solution was the fastest measured field operation. This matters when interpreting the value of the surrogate: in the homogeneous ADR1D problem, where a closed-form expression exists, the network offers no demonstrated computational advantage over that expression. Its value for integration must be evaluated in problems where the closed form is unavailable or where it forms part of a broader workflow. The current campaign does not yet measure that setting.

Peak memory traced from Python was approximately 27.6 MiB for ADR1D-ML loading and the full workflow, 25.9 MiB for the complete ADR1D-NN workflow, and 11.5 MiB for the complete numerical reference. These quantities exclude memory that native libraries do not expose to `tracemalloc`.

5 Discussion

5.1 What the Evaluation Demonstrates

The main strength of the evidence is not a single R^2 value but the sequence of controls: frozen models, a prior protocol, a collision-free challenge, one persisted inference, an audit from flat outputs, per-scenario comparison, and an independent traditional reference. Within that framework, ADR1D-ML retains the ability to recover effective parameters, and ADR1D-NN reproduces fields with low aggregate error.

For ADR1D-ML, velocity is the most stable and accurate output. Dispersion and decay exhibit greater relative variation, consistent with the difficulty of separating broadening, attenuation, and censoring from six noisy curves. The resolvability classifier discriminates well, but its precision of 0.5588 precludes treating a positive class as confirmation of decay. Its most reasonable use is to flag scenarios that require further estimation or additional information.

For ADR1D-NN, the absence of nonphysical values and exact satisfaction of imposed conditions are useful properties for integration into a numerical workflow. Even so, the hard constraints cover only specific parts of the problem. The maximum pointwise error and boundary profiles show that a network can satisfy its contract and retain a favorable global RMSE while locally displacing a front. Validation intended for maxima, arrival times, or concentration thresholds must retain local diagnostics.

The traditional campaign supports the analytical reference in the 16 cases and confirms that error decreases under refinement. The two intermesh exceedances occur at maximum Peclet. They do not invalidate the other comparisons, but they identify a specific condition under which medium resolution cannot be treated as equivalent to fine resolution under the adopted tolerance.

5.2 Joint Interpretation of the Models

Although ADR1D-ML and ADR1D-NN may occupy consecutive stages of a workflow, this campaign validated them separately. ADR1D-ML received sensor data and was compared with true parameters. ADR1D-NN received the true challenge parameters directly, not the estimates from

ADR1D-ML. The network RMSE therefore quantifies its surrogate error when its physical inputs are known; it does not include error from a preceding inverse problem.

This distinction has consequences. If a dispersion estimate with 25 % error is supplied to the network, the resulting field may differ from the solution computed with true dispersion even if ADR1D-NN approximates the supplied input perfectly. Likewise, a false positive in resolvability may introduce a positive λ in a case whose attenuation cannot be separated from noise. Errors from the two stages need not add linearly and may alter arrival times, amplitudes, and areas in different ways.

The current evidence consequently permits the interfaces to be integrated in controlled experiments, but does not permit the individual metrics of either model to be assigned to the chained workflow. An end-to-end validation would need to take the five sensor realizations, propagate the distribution of parameters estimated by ADR1D-ML into ADR1D-NN, and compare the resulting fields with the reference without replacing inputs by their true values.

5.3 What the Evaluation Does Not Demonstrate

The new scenarios belong to the same equation, geometry, and source family as the development benchmark. They are an independent sample within the domain, not a test of different physics. The validation also provides no field data in which hydraulic velocity, dispersion, porosity, and boundary conditions are observed simultaneously. Available WQP records provide context for concentrations, units, censoring, and provenance, but cannot by themselves reconstruct an identifiable case of Equation (1).

Bootstrap intervals quantify variation among scenarios in the design rather than predictive uncertainty for an individual case. ADR1D-NN is deterministic, and no ensembles, probabilistic calibration, or propagation of parameter uncertainty were evaluated. Spatial, temporal, and parametric extrapolation were likewise not tested.

The timing comparison does not isolate every implementation decision or represent different hardware. Loading a tree bundle and loading a neural checkpoint, in particular, do not have the same contract. Throughputs per second help size this execution but should not be generalized without repeating the benchmark in the target environment.

Table 10 summarizes the boundaries of interpretation.

Table 10: Claims supported and not supported by the campaign.

Question	Answer	Reason
Are the canonical evaluations reproduced?	Yes	The reconstructed reports were identical byte for byte.
Do the models work separately on new cases within ADR1D?	Yes, with limits	They met aggregate criteria but retained false positives and large local errors.
Is the ADR1D-ML → ADR1D-NN chain validated?	No	The network received true rather than inferred parameters.
Is use with field data validated?	No	Hydraulic parameters, boundary conditions, and an identifiable observational reference are missing.
Has extrapolation been demonstrated?	No	All cases belong to the ADR1D ranges, geometry, and equation.
Is there a universal speed-up?	No	Workloads, cardinalities, and computers are not equivalent; the analytical solution was also faster in this case.
Are the criteria equivalent to regulatory safety?	No	They are engineering tolerances specified for the synthetic benchmark.

5.4 Implications for Numerical Integration

The results support conditional integration rather than automatic replacement of the physical model. The following practices are recommended:

1. **Verify identity.** Checking the model, protocol, and inference modules against their manifests before loading an artifact avoids evaluating a version other than the documented one.
2. **Validate inputs.** Reject or flag positions, times, parameters, and pulses outside ADR1D ranges. A numerically valid output does not turn extrapolation into evidence.
3. **Retain replicates.** In the inverse problem, keep all five predictions and the resolvability probability, not only their means and classes, to represent sensitivity to noise.
4. **Propagate parameters.** If ADR1D-ML feeds ADR1D-NN, evaluate several plausible parameter sets instead of treating a point estimate as exact truth.
5. **Inspect the local field.** Accompany global RMSE with active-region error, arrival diagnostics, profiles near the source, and maxima. The two errors near 0.75 show why this step is needed.
6. **Maintain sentinel cases.** Solve boundary combinations and high-Peclet cases with an analytical or numerical reference for every version of the integrated workflow.
7. **Record cost by component.** Separate loading, transformation, and inference on the target hardware before choosing a deployment strategy.
8. **Revalidate when physics changes.** Heterogeneity, different boundaries, multidimensional geometry, or site data constitute a new problem and require a new protocol and new evidence.

In repeated simulations, ADR1D-NN may be useful as a field approximator and ADR1D-ML as a preliminary inverse stage. Neither replaces the assessment of identifiability, uncertainty propagation, or physical balance required by the final application.

6 Threats to Validity and Limitations

- **External validity.** The study is synthetic, one-dimensional, and homogeneous. It does not cover spatial heterogeneity, multidimensional transport, nonlinear reactions, or complete field data.
- **Physical independence.** The challenge uses a reconstructed implementation of the same analytical solution and the same physical ranges. Independence comes from the samples, locking, and audit rather than from a different transport theory.
- **Dependence between benchmark and models.** Both development and challenge labels come from ADR1D. The absence of collisions prevents repeated scenarios but does not remove shared assumptions about homogeneity, linearity, a rectangular pulse, and constant parameters.
- **Traditional reference.** Sixteen cases and three meshes were evaluated. This selection covers extremes and medoids but does not constitute an exhaustive convergence study.
- **Temporal resolution.** Arrival diagnostics use 1800 s steps in the challenge field; errors equal to that value are quantized by the evaluation table.
- **Noise.** The sensor model represents one combination of Gaussian error, an absolute floor, censoring, and substitution. It does not cover drift, temporal dependence, calibration bias, or sensor failures.
- **Conditional decay.** The λ regression is summarized from 23 resolvable scenarios. Its intervals are wider than those for velocity and must be retained when accuracy is communicated.
- **Definition of resolvability.** The threshold $1 - \exp(-\mathrm{Da}) \geq 0.03$ relates ideal attenuation to relative noise, but does not incorporate the absolute floor, censoring, or every correlation among parameters. It is an operational benchmark definition, not a universal limit of experimental detectability.
- **Unevaluated coupling.** ADR1D-NN received true parameters. The study does not quantify how ADR1D-ML errors and decisions propagate to the field.
- **Discontinuities.** The rectangular source changes instantaneously. The largest neural errors occur precisely near source activation and the first interior cell; other source forms may alter the error distribution.
- **Memory and time.** The benchmark was run on one computer. The Python memory probe may omit native allocations and is not equivalent to maximum resident memory.
- **Criteria.** The tolerances were specified before the challenge, but they are engineering criteria for this study rather than water-quality standards or guarantees for public decisions.

7 Reproducibility and Traceability

All results were stored in open CSV or JSON formats. The machine-readable protocol is located at `configs/validation_protocol.json`. The primary evidence is under `results/`, and generators and validators are under `src/`. Only `src/evaluate_locked_challenge.py` executed new inference on the challenge, once per model. Subsequent figure generation read persisted results exclusively and recorded `model_loading_or_inference = false`.

Reproducibility was organized into three scopes. *Integrity verification* confirms that every expected file exists and corresponds to its machine-readable record. The *results audit* recalculates metrics and intervals from persisted predictions without querying the models. A *complete reconstruction* regenerates cases or runs models and must be performed in a temporary directory so that a replica is not confused with the historical evidence. The final manifest uses the first two scopes and does not increment the challenge-inference counter.

Paths stored in the manifests are relative to the validation directory, so the collection can be moved to another computer without rewriting absolute paths. Each JSON records versions, seeds, counts, and integrity information for its inputs. Cryptographic fingerprints used by validators remain in those files and are not duplicated in the report.

Table 11: Primary traceability records and documented content.

Artifact	Status or content
`validation_protocol.json`	Locked protocol
`challenge_scenarios.csv`	120 scenarios
`challenge_analytical_field.csv`	299,880 values
`challenge_sensor_observations.csv`	176,400 observations
`canonical_reproduction.json`	Complete reproduction
`challenge_result_validation.json`	Passed audit
`numerical_reference_cases.csv`	48 runs
`numerical_reference_metrics.json`	Mixed evidence
`computational_cost.json`	11 measured workloads
`figure_manifest.json`	Eight graphical files

The four graphics are stored in both PDF and PNG. Two independent temporary renderings produced identical files in all eight formats. The manifest records the inputs, generator, and each figure, as well as the criterion that selected `A5-CH-0056` for field inspection.

The cost audit verified 14 artifacts before and after execution, with no changes, and recorded zero challenge inferences and zero newly persisted predictions. The cost report also reproduced the canonical results of both models and the numerical-reference RMSE values within tolerances below 10^{-12}.

The report itself is also part of traceability. Its internal date and PDF trailer identifier were fixed before compilation; two builds in separate directories must produce the same binary file. This condition verifies that the delivered document corresponds to the LaTeX source and registered figures, not that the scientific results are correct merely because it compiles.

8 Conclusions

The validation supports seven bounded conclusions:

1. ADR1D-ML reproduced its canonical results and met every challenge criterion. Velocity estimation was the strongest output; dispersion and decay retained greater relative uncertainty.
2. The resolvability classification distinguished the classes well in aggregate but produced 15 false positives and a precision of 0.5588. Probability and use context must accompany any positive class.
3. ADR1D-NN achieved good global fit, preserved the physical interval, and imposed the hard conditions exactly. Maximum errors near 0.75 and the worst RMSE of 0.05323 show that a global summary does not replace analysis of fronts and boundary profiles.
4. The traditional solution agreed with the analytical solution and improved monotonically in all 16 cases. Two scenarios with Peclet 350 did not meet the medium–fine tolerance; by design, the aggregate numerical result is mixed evidence.

5. The observed times show that both models can be evaluated in fractions of a second for the tested workloads. Differences in task, mesh, and cardinality prevent converting these measurements into a universal speed-up.

6. The two models were validated separately. ADR1D-NN received true parameters, so its metrics do not describe a workflow fed by ADR1D-ML. Propagation of inverse error into the field remains to be evaluated specifically.

7. The models are ready for integration into controlled numerical experiments within the ADR1D domain, with input verification, retention of replicates, and physical checks. The evidence does not authorize extrapolation, field validation, or regulatory use without an additional campaign.

The immediate next step is to evaluate the complete chain within the same domain: noisy sensors, inferred parameters, neural field, and final comparison with the reference. Subsequent work can examine how the evidence changes when heterogeneity, different boundary conditions, and sufficient observations to identify the physical problem are introduced. Every extension should retain the same order: a prior protocol, frozen artifacts, local metrics, and publication of both favorable and unfavorable results.

Declarations

Author contributions. Gerardo Tinoco-Guerrero, Francisco J. Domínguez-Mota, and J. Alberto Guzmán-Torres contributed equally to the design, analysis, interpretation, and review of the work.

Conflicts of interest. The authors declare no conflicts of interest.

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Data and code availability. The original ADR1D data are available at `https://github.com/gstinoco/ADR1D`. The models are published at `https://github.com/gstinoco/ADR1D-ML` and `https://github.com/gstinoco/ADR1D-NN`. Evidence specific to this validation is preserved with this report in readable and auditable CSV, JSON, PDF, and Python files.

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